Research update after supervisor feedback

Color constancy inside geometry estimation

Shirsha Mitra / Student ID 2023521460130 / 29 October 2026

This phase acquires matched multi-light data, measures lighting sensitivity, and implements a small CC-inspired module inside a depth model. It preserves the first NYU preprocessing study as a separate baseline.

Completed action	Evidence
Dataset search and acquisition	Vari: 528 RGB images, 48 fixed-camera groups, 11 lighting conditions, 2 rendered rooms. DTU: 42 real RGB images, 2 scenes, calibrated cameras and structured-light reference surfaces.
Task-integrated prototype	101,504 trainable parameters between the frozen DINOv2 encoder and frozen depth decoder. Raw RGB remains unchanged. Depth and paired-light consistency train the internal features.
Controlled comparisons	Frozen model, image preprocessing, ordinary task adaptation, achromatic conditioning, chromatic conditioning, and consistency ablation.
Current interpretation	Lighting affects geometry. The present chromatic module does not outperform ordinary task adaptation; a CC-specific gain is not established.

Same camera and geometry, different rendered lighting

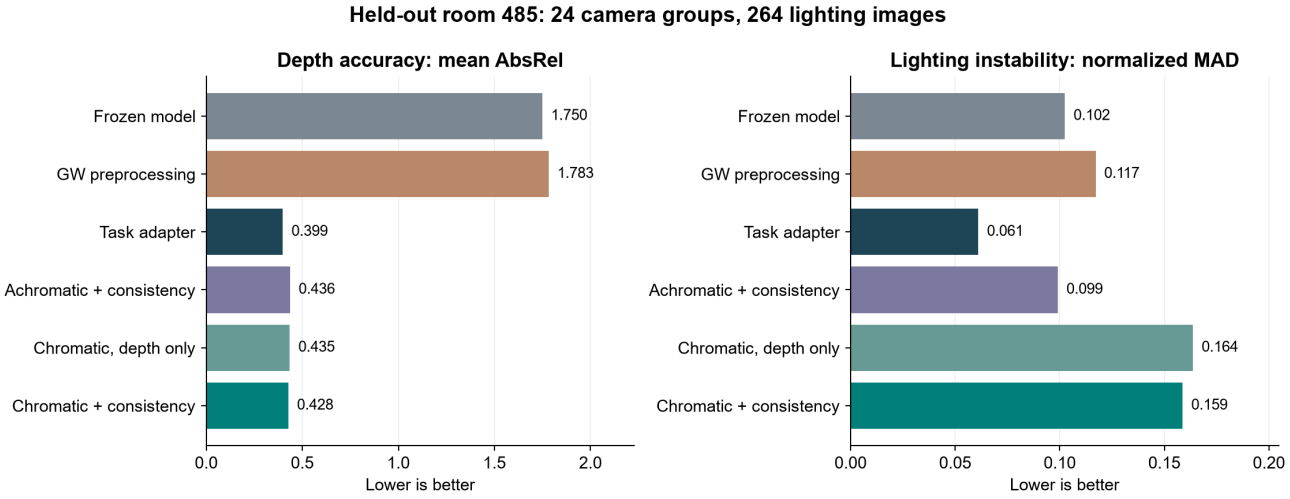


Data limits: Vari publicly releases only its test archive; we create a new exploratory split, with room 212 for development and room 485 for evaluation. These are not official benchmark scores. DTU varies lighting direction and does not supply ground-truth illuminant colors.

Monocular depth experiment

Development: 20 training and 4 validation camera groups in room 212. Evaluation: 24 camera groups in room 485, 264 lighting images. The evaluation contains one unseen room, not 24 independent scenes. All methods use the same 224-pixel input transform, reference pixels and frozen depth backbone.

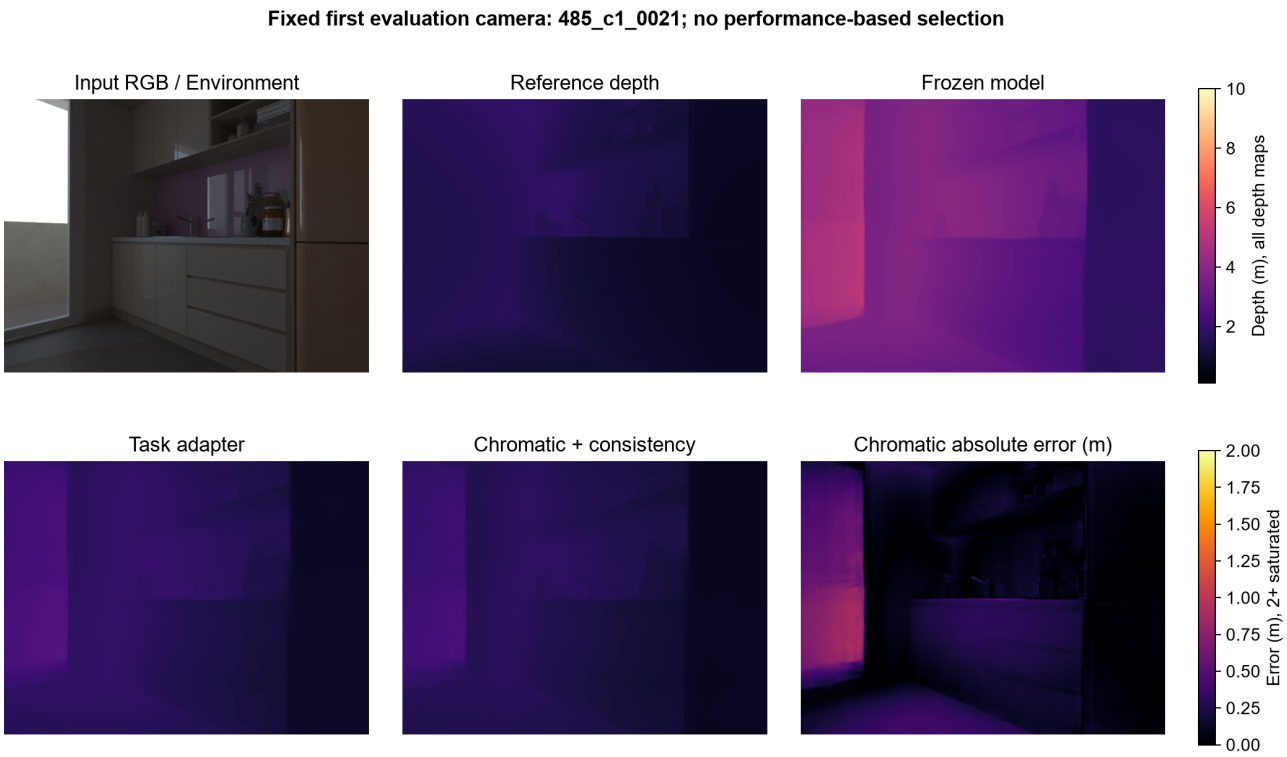
Method	AbsRel	RMSE (m)	Delta1	Light MAD
Frozen original	1.7497	2.982	0.2%	0.1025
Gray-World preprocessing	1.7827	3.049	0.2%	0.1171
Task adapter (no color cue)	0.3988	0.707	25.6%	0.0611
Achromatic + consistency	0.4360	0.782	24.0%	0.0991
Chromatic / depth only	0.4347	0.770	29.2%	0.1637
Chromatic + consistency	0.4277	0.758	29.7%	0.1587



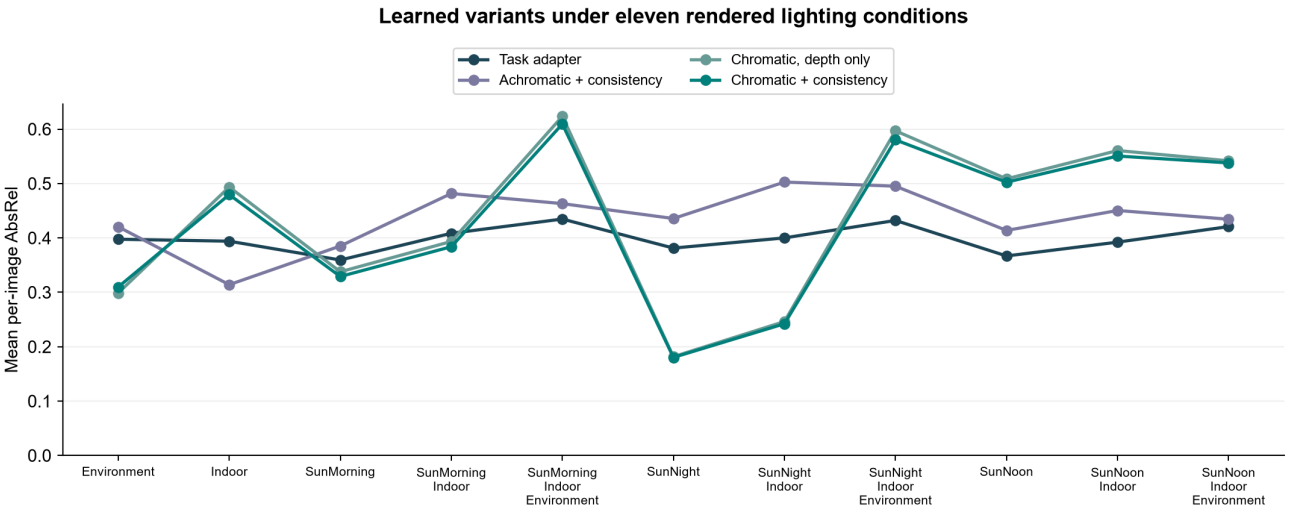
The chromatic consistency model has 7.2% higher AbsRel than the ordinary task adapter (0.4277 versus 0.3988). Its large improvement over the unadapted model is therefore not evidence that CC itself caused the gain.

AbsRel and RMSE are means of per-image scores. Light MAD is the mean absolute deviation across eleven lighting predictions around their per-pixel mean, divided by reference depth. Lower is better for these three errors; higher Delta1 is better. Accuracy and stability must be interpreted together.

Saved depth predictions and ablations



The example is the first sorted evaluation camera and lighting condition, chosen without ranking model performance. All depth maps use the same 0.1-10 m color scale. The error map saturates at 2 m. The ordinary adapter is an essential control for general domain/scale adaptation.



Training uses 150 paired updates per learned variant, AdamW and train-only descriptor statistics. Validation selects checkpoints only in room 212. Primary references preserve sixteen-bit TIFF depth; the authors-compatible eight-bit path is retained and checked separately. No prediction-specific scale fitting occurs.

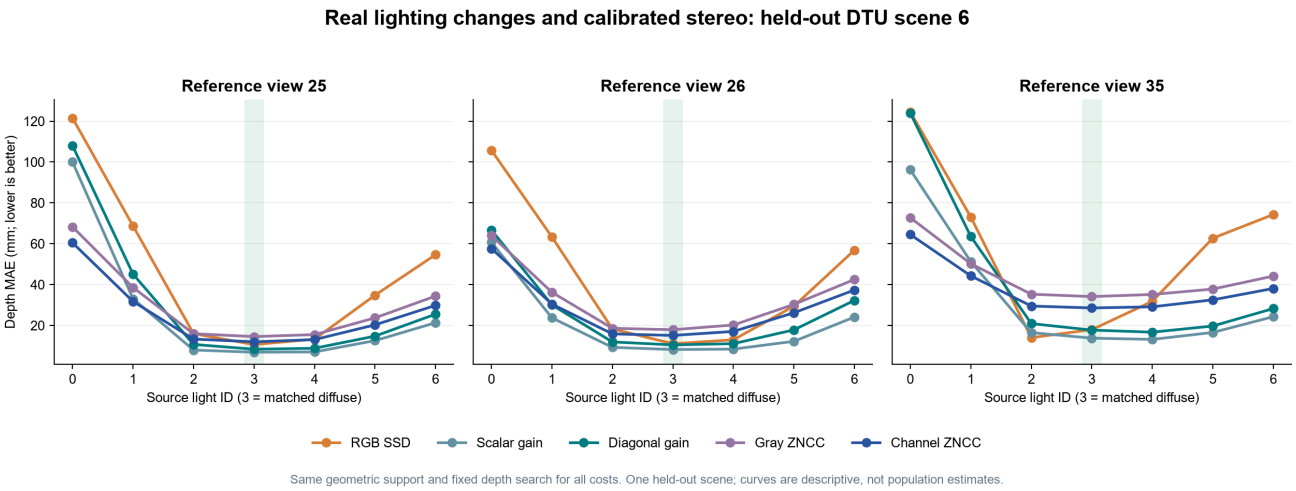
Calibrated real-lighting geometry study

DTU uses real images and original structured-light surfaces. The same calibrated view pair is tested under matched lighting and a changed source light. Plane-sweep matching estimates depth directly; radiometric gains are fitted inside each depth-hypothesis cost, rather than writing corrected RGB images.

The final stereo results and numerical checks are available in phase2/results/stereo/main_v2. Methods include RGB SSD, scalar gain, diagonal gain, grayscale ZNCC and per-channel ZNCC. The known robust costs are strong baselines; the gain-based prototype is not a novelty claim.

Method	Matched MAE (mm)	Changed-light MAE (mm)	Ratio
RGB SSD	13.19	54.21	4.11x
Scalar gain	9.61	29.90	3.11x
Diagonal gain	12.17	36.43	2.99x
Gray ZNCC	22.18	37.97	1.71x
Channel ZNCC	18.54	32.84	1.77x

On held-out scene 6, raw RGB matching increases from 13.19 mm MAE under matched lighting to 54.21 mm under changed source lighting (4.11 times). Modeling radiometric nuisance inside matching helps, but the diagonal-channel candidate (36.43 mm) is worse than scalar gain (29.90 mm) and established channel ZNCC (32.84 mm). This supports lighting sensitivity in geometry; it does not establish an extra benefit from color constancy.



Units are millimeters. Scene 1 supports geometry checks; scene 6 is held out. Masks and depth planes are shared across methods. Reference surfaces enter evaluation and visibility checking, never matching costs. One held-out scene cannot support a population-level confidence claim.

What this phase establishes

Completed: supporting datasets were acquired and audited; geometry was tested under matched lighting variants; a trainable CC-inspired feature module now participates inside depth inference; strong task-adaptation and lighting-invariance controls were evaluated.

Remaining gap: this prototype has not demonstrated an advantage attributable specifically to CC. The unadapted model has a large domain/scale bias on Vari, and the ordinary task adapter remains stronger than the current chromatic version. Two rooms are insufficient for broad learning claims.

Next research decision: obtain more diverse scene-separated paired-light data; test cues less dependent on average room color, such as neutral-surface or relative-color evidence; retain achromatic and ordinary-adaptation controls. Improve the method on development data before evaluating another independent scene set.

Reproduction and evidence

Project: D:/thesis work/thesis code
Branch: research/phase2-integrated-cc
GitHub: https://github.com/shirsha2002/color-constancy-depth-thesis/tree/research/phase2-integrated-cc

Fresh monocular reproduction:
.venv/Scripts/python.exe phase2/audit/reproduce_guard.py --run-fresh --out phase2/results/mono_reproduction_01
The destination must not exist. The guard verifies frozen inputs and prevents stale cache reuse. See phase2/README.md for data acquisition, audits and stereo settings.

Depth Anything V2 metric implementation
https://github.com/DepthAnything/Depth-Anything-V2/tree/main/metric_depth

Vari dataset and DCA authors
https://github.com/zzt76/dca

DTU original data and calibration
https://roboimagedata.compute.dtu.dk/?page_id=36

FiLM feature modulation background
https://arxiv.org/abs/1709.07871